



Chihyeon Yun

AI Researcher & Engineer

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- ◆ **Multimodal Auto-Editing (M.S. & First Author):** Designed visual speaking-state (Swin Transformer) and audio diarization (WhisperX / Pyannote) late-fusion frameworks, utilizing Beam Search decoding to estimate optimal camera cuts (achieved 90.10% agreement with experts).
- ◆ **Spatial AI & Metaverse (ETRI R&D):** Engineered high-accuracy indoor Visual Positioning Systems (VPS) for cyber-physical space registration, and designed Visual-Language Model (VLM) zero-shot anomaly detection pipelines for real-time video semantics.
- ◆ **Acoustic Surveillance (Scientific Data, Nature SCIE):** Built the first open Military Audio Dataset (MAD, 8,075 samples) for situational awareness, designed log-Mel filterbank feature extractions, and benchmarked Audio Spectrogram Transformers (AST, 91.07% accuracy).
- ◆ **Medical AI & Audio Diffusion (NeurIPS Workshop):** Addressed diagnostic data scarcity via conditional audio diffusion (DiffWave) synthesis and designed Adversarial Fine-Tuning (AFT) models to align domain features (improving minority class classification by up to 26.58%).
- ◆ **3D Reconstruction & VLA Robotics (PoC):** Coupled 2D MT-DUNet (DCNv2) and Beer-Lambert radiative 3DGS for coronary artery 3D reconstruction under extreme memory optimization, and integrated SmolVLA control in Unity ROS simulation environments.
- ◆ **Multimodal Emotion Recognition (KCC & ETRI Award):** Integrated text (KLUE-RoBERTa) and speech (wav2vec 2.0) encoders using Latent CutMix augmentation and Supervised Contrastive Learning (SupCon) for cosine similarity alignment (F1-score 71.17%).

📁 Research & Career Timeline

2025.02 - 2025.06

Electronics and Telecommunications Research Institute (ETRI)

Post-Master's Research Fellow at Hyper-Realistic Metaverse Lab. Worked on indoor visual positioning (VPS) coordinates mapping and VLM semantic anomaly detection. Managed joint government R&D milestones.

2023.03 - 2025.02

Kyungpook National University

Master of Science in Artificial Intelligence (GPA: 4.11/4.3). Conducted research in Multimodal Language & Cognition Lab on audio-visual late-fusion, speech diarization, and generative audio diffusion.

🔧 Core Tech Stack & Skills

Deep Learning & Audio

PyTorch Transformers DiffWave Ko-wav2vec 2.0 KLUE-RoBERTa
SupCon WhisperX Pyannote.audio Librosa

Vision & Spatial AI

3DGS Visual SLAM/VPS MT-DUNet (DCNv2) Swin Transformer OpenCV
MediaPipe

VLMs, Robotics & DevOps

VLA / SmolVLA VLM Zero-shot ROS/ROS2 Unity Engine Docker
Pinecone Vercel Git

Medical Imaging

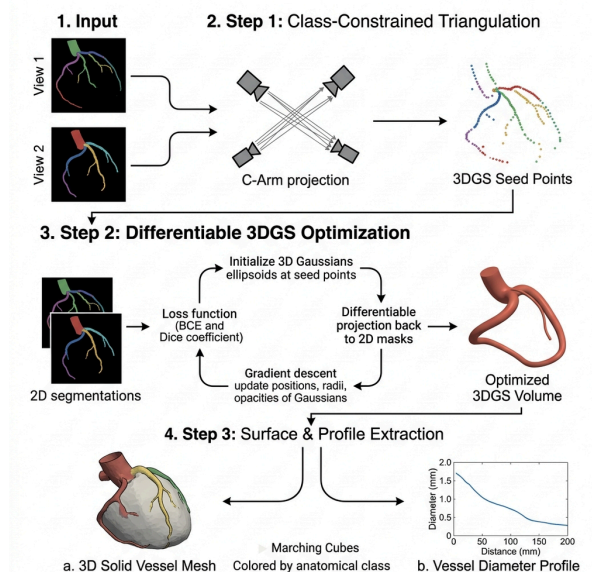
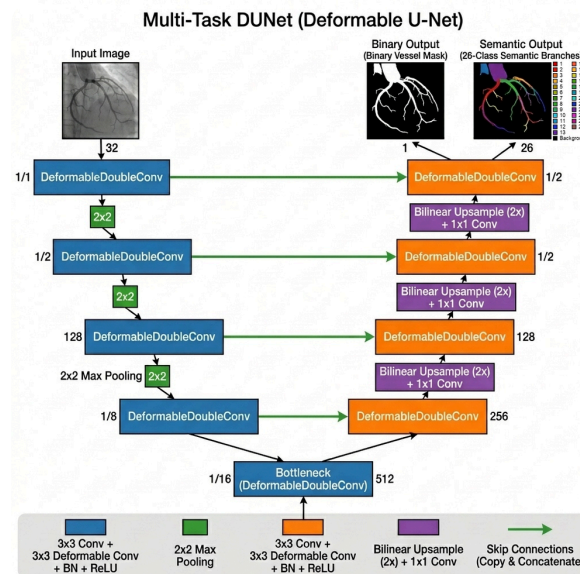
3D Reconstruction

LCA CAG 3DGS Reconstruction & Dashboard

A 3D volumetric reconstruction PoC that combines a 2D Multi-Task Deformable UNet (MT-DUNet) and Beer-Lambert radiative 3D Gaussian Splatting (3DGS) to recover 3D coronary artery structure, radius profile, and volume from sparse 2D C-arm X-ray angiograms.

- ✦ **MT-DUNet (DCNv2):** Integrates deformable convolutions to adapt to complex vessel tortuosity, simultaneously predicting binary vessel masks and 25 anatomical branch labels.
- ✦ **Class-Constrained Triangulation:** Prevents ghost-point artifacts in epipolar matching by pairing identical anatomical branch labels across views, yielding a 4.5x denser centerline seed.
- ✦ **Beer-Lambert Radiative 3DGS:** Solves the transparency occlusion problem of X-ray projection by analytically integrating density along rays, optimizing vessel radius (r) and opacity (α).
- ✦ **High Efficiency:** Achieved 8x GPU memory saving (2.5 GiB execution) on an L4 GPU via downsampling and multi-resolution rendering.

Tech Stack: PyTorch, 3DGS, Deformable Conv (DCNv2), Python, HTML/JS



AI Agent

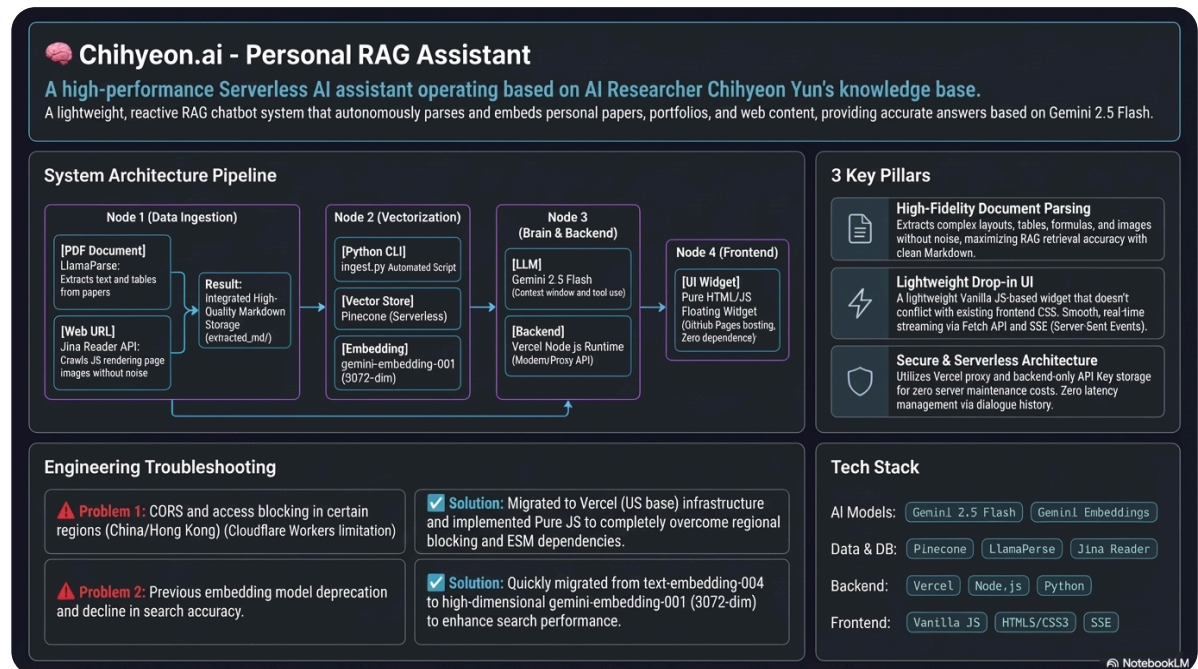
RAG Systems

Chihyeon.ai Personal RAG Assistant

A sophisticated personal RAG system that serves as an interactive AI secretary, providing high-fidelity, grounded answers based on academic research papers, CV, and project documentation.

- ◆ **High-Fidelity Parsing:** Leverages LlamaParse & Jina Reader API to flawlessly extract complex academic layouts, mathematical formulas, and JS-rendered web content.
- ◆ **Secure Serverless Proxy:** Uses Vercel Functions to encrypt API credentials and bypass region locks, ensuring safety and compliance.
- ◆ **Vector Storage:** Fast semantic matching over high-dim Pinecone space powered by Gemini 2.5 Flash for near-instant, streaming answers.

Tech Stack: Gemini 2.5 Flash, Pinecone (Serverless), Vercel Functions, LlamaParse, Jina Reader, Python



Robotics

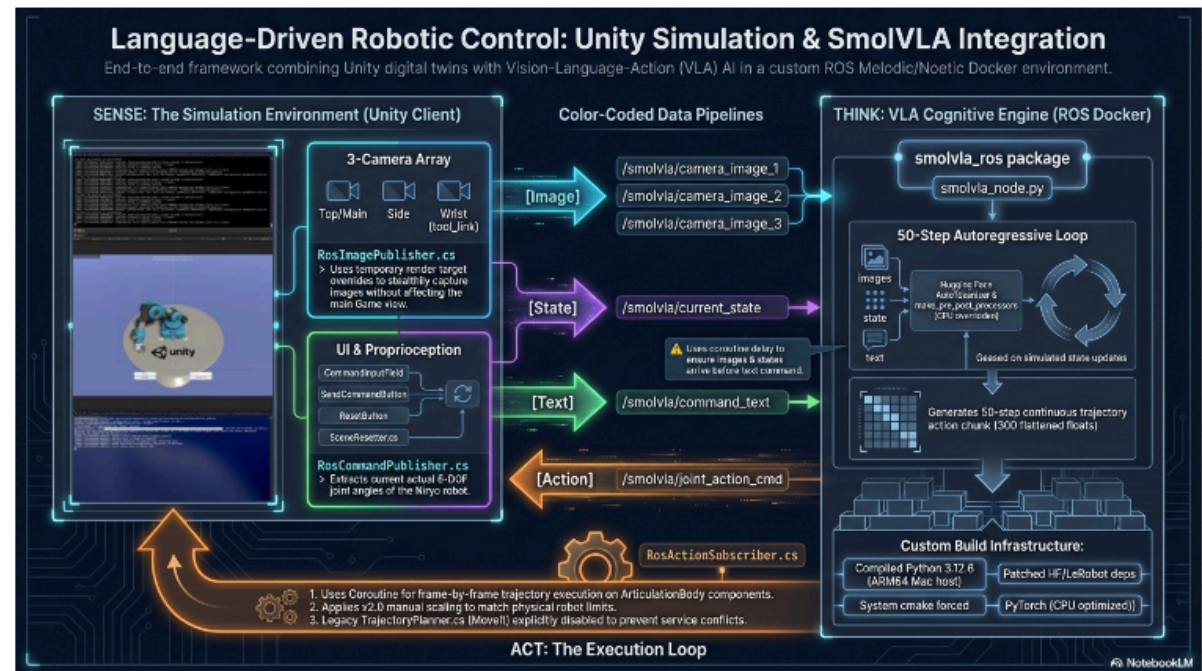
Simulation

Unity Pick & Place SmolVLA Integration

An end-to-end integration of the SmolVLA (Vision-Language-Action) model within a virtual physics environment to perform robotic 'Pick and Place' tasks.

- ◆ **Vibe Coding Paradigm:** Worked with LLMs to prototype network interfacing, setup boilerplate code, allowing focus on core logic.
- ◆ **Communication Pipeline:** Built ROS-Unity bridges linking virtual cameras and joint torques for real-time simulation feedback.
- ◆ **Visual Control Loop:** Feeds camera images directly to SmolVLA agents for high-level spatial coordination and torque control verification.

Tech Stack: SmolVLA, Multimodal LLM, Unity Engine, ROS, Docker, PyTorch



Seamless Multi-Hypothesis Video Editing Using Deep Multimodal Analysis

Fuses active speaker detection (Swin Transformer) and diarization (WhisperX/Pyannote) visual-audio streams to align channels, decoding edits via a customized Beam Search logic matching 90.10% of expert visual pacing.

- Proposed a late-fusion multi-camera synchronization and editing framework, integrating speech state classification and visual cue matching into a Beam Search sequence decoder.

Tech Stack: Speaker State Estimation, Swin Transformer, WhisperX, Pyannote.audio, Beam Search, Python

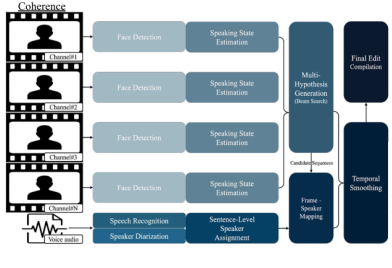
Seamless Multi-Hypothesis Video Editing Using Deep Multimodal Analysis Over Multi-Channel Sources

Introduction

- An automated system for editing multi-channel video by synchronizing raw inputs through separate visual and auditory pipelines.
- Deep late-fusion of multimodal information for face recognition, speaking-state estimation, speech recognition and speaker identification.
- A multi-hypothesis editing approach that provides candidate versions based on probability distribution over time for extracting frames from each channel stream.
- Frame-level match accuracies of 84.99% and 90.10% for the 3-channel and 5 channel videos, highlighting the system's robust performance across varying speaker complexities.

Method

- For multi-speaker, multi-channel content, we adopted an **intuitive approach** to make automated editing feasible. This is primarily because **datasets containing multi-speaker, multi-channel videos paired with their fully edited versions are extremely rare**.
- The lack of such datasets makes it challenging to apply standard supervised learning techniques effectively for automated video editing.
- For the audio modality, our goal is to accurately detect active speech segments and identify the corresponding speakers from diverse and often noisy multi-channel sources - **Speech Recognition, Speaker Diarization**
- The visual approach involves estimating the speaking state based on the facial region of the speaker - **Face Detection, Speaking State Estimation**
- Integration and Channel Selection - **Multi-Hypothesis Generation for Temporal Coherence**



Experiment & Result

- Dataset
 - Studio recordings provided by a Korean public broadcasting company in South Korea. {3-channel video dataset, 5-channel video dataset}
- Evaluation
 - Compare the ground-truth labeled data with output generated by our system at the frame level.
 - For the Quantitative Analysis, Analyze the frame-level alignment accuracy under various smoothing and temporal conditions.

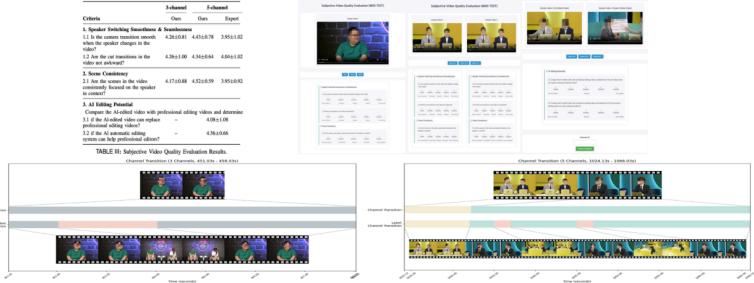
Duration	Thr 15	Thr 30	Thr 45
<i>Best Path (self-transition bias 0.7)</i>			
3-channel video	0.8498	0.8495	0.8495
5-channel video	0.8995	0.8999	0.8854
<i>Best Path (self-transition bias 0.9)</i>			
3-channel video	0.8499	0.8497	0.8481
5-channel video	0.8987	0.9010	0.8865

TABLE I: Match rates (Top Beam Paths) applying self-tracing (0.7/0.9) varying the smooth threshold.

Length (s)	3-channel video			5-channel video		
	Match	Frames	Rate	Match	Frames	Rate
0-5	3,849	6,660	0.5779	2,981	4,335	0.6876
5-10	4,162	4,065	0.8885	5,447	5,940	0.9169
10-15	5,141	5,065	0.9665	7,867	8,225	0.9442
15-20	3,917	3,210	0.9525	3,573	3,780	0.9452
20-30	1,140	1,140	1.0000	842	850	0.9898
30-40			0.0000	2,124	2,580	0.9898
40-50			0.0000	2,432	2,850	0.9898
50-60	3,104	3,315	0.9363	1,648	1,950	0.9898
60-70			0.0000	2,010	2,010	1.0000
70-80	2,128	2,130	0.9991	2,198	2,220	0.9939
80-90			0.0000	2,670	2,670	1.0000
Total	22,515	26,490	0.8499	34,330	38,100	0.9009

TABLE II: Match rates across different segment lengths.

- For the Qualitative Analysis, Analyze subjective assessments of editing quality and viewer experience.
- The analysis was performed with 24 participants, including developers and broadcasting industry professionals, under a blind setup, where they were not informed whether a video was edited by our system or by a human expert.



02 Scientific Data (Nature Portfolio, SCIE, 2024 / Co-Author)

A Military Audio Dataset for Situational Awareness and Surveillance

Curated and open-sourced 8,075 military audio clips (MAD) representing 7 hazard classes for security surveillance, designing raw audio preprocessing filters and evaluating AST models achieving 91.07% classification accuracy.

- ✦ Built audio feature extraction pipelines (Librosa log-Mel filterbanks) and structured AST-Patch-Mix classifier benchmarks for real-time audio defense surveillance.

Tech Stack: Audio Spectrogram Transformer (AST), Mel Filterbank, Python, Librosa

A Military Audio Dataset for Situational Awareness and Surveillance

Introduction

- Audio classification related to military activities is a challenging task due to the high levels of background noise and the lack of suitable and publicly available datasets.
- To bridge this gap, this paper constructs and introduces a new military audio dataset, named MAD, which is suitable for training and evaluating audio classification systems.
- The proposed MAD dataset is extracted from various military videos and contains 8,075 sound samples from 7 classes corresponding to approximately 12 hours, exhibiting distinctive characteristics not presented in academic datasets typically used for machine learning research.
- We further conduct a comprehensive sound classification study of various deep learning algorithms on the MAD dataset.

Method

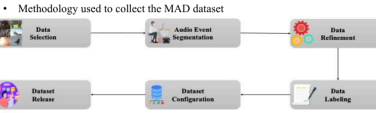


Fig. 1 Illustration of MAD dataset collection procedure.

- Data selection
 - Identify the audio content types for the seven military-related categories.
- Audio event segmentation
 - Identify the start and end times of each audio event.
- Data refinement
 - Refine the data using a variety of techniques, such as formatting, resampling, and extracting.
- Data labeling
 - Label all audio events into one of the seven predefined classes.
- Data configuration
 - Randomly split all the data into train/test sets with an approximately 9:1 ratio.

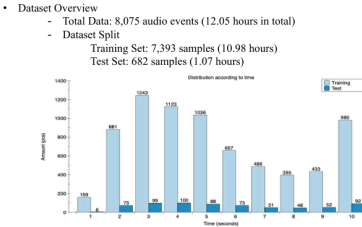


Fig. 2 Bar chart illustrating the distribution of training and test sets according to various times.

- Key Features
 - All audio events range from 1 to 10 seconds in duration. Meticulously labeled and categorized for high-quality deep learning-based audio classification tasks.
- Total Classes: 7 audio classes, including Communication, Gunshot, Footsteps, Shelling, Vehicle, Helicopter, and Fighter.

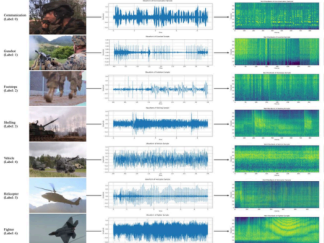


Fig. 4 The overall pre-processing task for converting waveforms to Mel filterbanks.

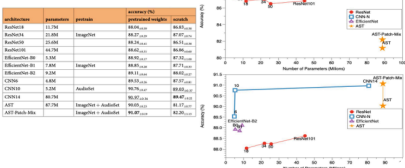
Technical Validation

- Model Evaluation: We conducted a comprehensive analysis of various deep learning models to evaluate the performance of the MAD dataset.
 - Models: ResNet, EfficientNet, CNN-N (PANNs), and AST (Audio Spectrogram Transformer).
- Pre-processing
 - Audio waveforms were converted into Log Mel filterbank features (80 dimensions, 25ms window, 10ms overlap).
 - All audio samples were standardized to a 10-second length using zero padding.
 - Standard normalization was applied to the spectrograms.
 - Augmentation: SpecAugment was used to mitigate overfitting.



Fig. 5 Illustration of overall deep learning-based audio classification process.

- Experimental Result Summary
 - Pretrained models consistently outperformed those trained from scratch. With pretrained weights, the AST-Patch-Mix model achieved the highest accuracy (91.07%), aligning with the trend that models with more parameters perform better. A notable exception was the CNN10 model, which achieved a strong 89.53% accuracy with only 6% of the parameters of the AST model.
 - In contrast, when trained from scratch, the CNN14 model performed best (89.47%), while the high-parameter AST models surprisingly showed the lowest performance.



03

NeurIPS 2023 DGM4H Workshop (International Conference / Co-Author)

Adversarial Fine-tuning using Generated Respiratory Sound

Tackled medical data scarcity via DiffWave diffusion synthesis, training deep classification structures with domain-adversarial fine-tuning (AFT) loops to bridge synthetic and real sample sets, increasing minority class recall by 26.58%.

- Designed medical sound synthesis pipelines using conditional audio diffusion neural vocoders, and proposed domain-adversarial fine-tuning for diagnostic label alignment.

Tech Stack: DiffWave, Domain Adversarial Training (DAT), Audio Synthesis, PyTorch



Adversarial Fine-tuning using Generated Respiratory Sound to Address Class Imbalance

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Introduction

- While *deep generative models* have emerged as a promising approach in the medical image domains to address data scarcity, their use for sequential data like **respiratory sounds** is less explored.
- In this work, we propose a straightforward approach to augment imbalanced respiratory sound data using an **audio diffusion model** as a **conditional neural vocoder**.
- Nevertheless, the **discrepancy between synthetic and real samples** can introduce problems related to **distribution inconsistency**, which can degrade the performance of respiratory sound classification models as the proportion of synthetic data in the training set increases.
- We further introduce a simple yet effective **Adversarial Fine-tuning** method to **align features between the synthetic and real respiratory sound samples** to improve respiratory sound classification performance.

Method 1

- We aim to generate **high-fidelity respiratory sound samples** using audio diffusion model [1] and then **combine these synthetic samples with real data to improve the respiratory sound classification task**, especially for imbalanced lung sound disease classes.
- In phase 1, we introduce a simple method for **generating respiratory sound samples** using a **conditional neural vocoder** inspired by the recent success of *DiffWave* [1] in obtaining realistic audios.

Phase 1: Respiratory Sound Generation

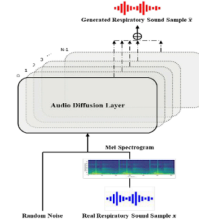


Figure 1: Overview of respiratory sound generation pipeline using audio diffusion model.

Method 2

- While augmenting real data with synthetic samples can be beneficial, we found in early experiments that in our case the **distribution mismatch between the two types of data degraded the performance of the classification model**.
- To overcome this issue, we propose a simple yet effective **Adversarial Fine-Tuning (AFT)** method inspired by [2].

Phase 2: Adversarial Fine-tuning with Mixed Samples for Respiratory Sound Classification

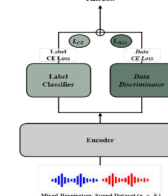


Figure 2: Overview of Adversarial fine-tuning with synthetic and real samples.

- The proposed method consists of two losses with label classifier L_{CE} and data discriminator L_D :

$$L_{CE} = -\frac{1}{N} \sum_{i=1}^N y_i \log(\hat{y}_i), \quad L_D = -\frac{1}{N} \sum_{i=1}^N d_i \log(\hat{d}_i),$$

- where L_{CE} and L_D are CE loss with label y and label d , and the predicted probabilities $\hat{y}$ and $\hat{d}$ are obtained by passing through the classifier and discriminator, respectively.
- To ensure that the **learned features cannot distinguish between the synthetic and real samples**, **gradients from L_D are multiplied by a negative constant during the backpropagation**.
- The final training objective is $L_{AFT} = L_{CE} + \lambda L_D$, where λ is a regularization parameter drawn from [2].
- We used Audio Spectrogram Transformer (AST) [3] model as our baseline.

Results

Table 1: Performance according to various mixed sample amounts.

method	Net Avg.	Mixed 50%	Mixed 60%	Mixed 70%	Mixed 80%	Mixed 90%	Mixed 95%
AST FT	59.55	59.92 _(+0.37)	59.99 _(+0.07)	59.81 _(-0.18)	59.65 _(-0.16)	59.18 _(-0.47)	58.56 _(-0.62)
AFT	—	61.79 _(+2.27)	60.89 _(-0.90)	60.81 _(-0.08)	60.03 _(-0.78)	59.08 _(-1.05)	59.74 _(+0.66)

Table 2: Accuracy (%) of the abnormal classes on the C201 test set for AST FT and AFT.

class	ratio	AST FT (Net Avg.)	AST FT (Mixed 50%)	AST FT (Mixed 60%)	Adversarial FT (Mixed 50%)	Adversarial FT (Mixed 70%)
crackle (C ₁)	23.55%	45.45	42.84	42.86	41.07	46.99
wheezes (C ₂)	1.97%	36.62	36.1	22.08	27.92	36.12
both (C ₃)	5.19%	13.38	9.09	7.69	41.98	33.66

Table 3: Overall comparison of the IC-BIH dataset for the respiratory sound classification task.

method	architecture	pretrain	S_{CE} (%)	S_{CE} (%)	Score (%)
ResNet18 [14] (IC-BIA+BBC+FT)	ResNet18	IN	72.30	40.10	56.30
Wang et al. [12] (DiffNet)	ResNet18	IN	70.40	40.20	55.50
Nguyen et al. [23] (CoTraining)	ResNet18	IN	79.34	37.24	58.29
Mohamed et al. [21] (DCL)	CNN6	AS	75.91	39.15	57.55
Bae et al. [11] (Diffusion)	AST	IN + AS	77.14	41.97	59.55
Bae et al. [11] (Diffusion)	AST	IN + AS	81.66	43.97	62.97*
AFT on Mixed-50% [ours]	AST	IN + AS	80.22 _(+1.44)	42.85 _(-1.1)	61.79 _(+0.82)

- While directly fine-tuning on mixed data did not improve the performance of the minority classes overall, **our proposed method improved their accuracies by up to 26.58%, especially in C₂**.
- Furthermore, our proposed method trained with **Mixed-50%** achieved a score of **61.79%**, **outperforming the AST FT model by 2.24%.**

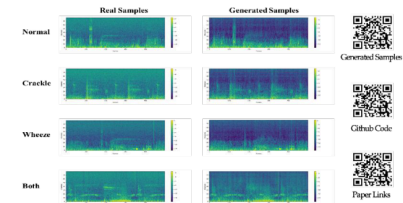


Figure 3: Comparison of spectrograms per class randomly chosen from the test set and the generated results.

References

- [1] Kang Zhifeng, et al (2021), "DiffWave: A versatile Diffusion Model for Audio Synthesis", 2021.10.01
- [2] Guo Yuxuan, et al (2018), "Domain-Adversarial Training of Neural Networks", The Journal of Machine Learning Research 2018
- [3] Gong Yoon, et al (2021), "AST: Audio Spectrogram Transformer", INTERSPEECH 2021

04

KCC Proceedings (Domestic Conference / Bronze Prize)

CutMix-SupCon: Contrastive Learning for Multimodal Emotion Recognition

Designed latent-level CutMix feature interpolation to merge KLUE-RoBERTa (text) and wav2vec 2.0 (audio) embeddings, optimizing Supervised Contrastive Learning (SupCon) for cosine similarity alignment on minor classes.

- Implemented multimodal late-fusion and contrastive learning alignments, improving speaker emotion recognition rates on class-imbalanced benchmark sets.

Tech Stack: wav2vec 2.0, KLUE-RoBERTa, Supervised Contrastive (SupCon), CutMix

CutMix-SupCon: Supervised Contrastive Learning with CutMix Augmentation for Multimodal-based Emotion Recognition

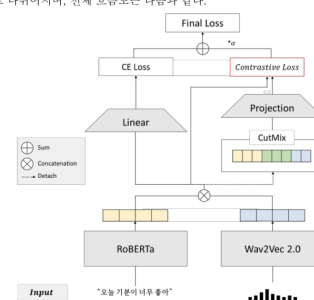
서론

- 멀티모달 데이터는 구조적으로 획득하기 어려운 문제가 있으며, 부족한 데이터 샘플의 문제로 인해 학습된 딥러닝 기반 네트워크의 실용화가 진행되기에는 여전히 한계점이 존재한다.
- 선행 연구들은 실제 환경의 부족한 데이터 문제는 고려하지 않고, 멀티모달 결합 관점에 대해서만 집중되는 경향이 존재한다.
- 본 연구에서는 멀티모달 데이터셋의 불균분한 문제를 해결하기 위해 CutMix-SupCon 방법을 제안한다. 제안된 CutMix-SupCon 방법은 잠재 공간에서 추출된 음성-텍스트 모달 간의 결합된 특징표현으로부터 혼합된 특징표현을 만들고, 혼합 대표 손실을 통해 음성-텍스트의 결합된 멀티모달 특징 표현이 원본과 얼마나 유사한지에 대한 목적 함수를 통해 최적화하였다.

방법

제안하는 멀티모달 CutMix-SupCon 방법은

- 멀티모달 특징표현 결합
 - CutMix 기반 멀티모달 혼합 증강
 - SupCon
- 으로 나뉘어지며, 전체 흐름도는 다음과 같다.



- CutMix 기반 멀티모달 혼합 증강
멀티모달 특징표현 $\mathbf{z}$ 과 레이블 y 에 대하여 샘플 $(\mathbf{x}_A, y_A)$, $(\mathbf{x}_B, y_B)$ 를 구성할 수 있고, 이로부터 새로운 샘플 $(\mathbf{z}, y)$ 를 아래의 식과 같이 만들 수 있다.

$$\mathbf{z} = M \odot \mathbf{x}_A + (1 - M) \odot \mathbf{x}_B$$

$$y = \lambda y_A + (1 - \lambda) y_B$$

- CutMix-SupCon
CutMix 기법으로부터 증강된 멀티모달 특징표현을 위한 적절한 손실 함수를 적용하기 위해, 잠재 공간내에서 혼합된 정보를 구분하여 효과적인 학습을 가능하게 하는 CutMix-SupCon 방법을 제안한다.

$$L_{CL} = \sum_{i \in I} \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\mathbf{z}_i \cdot \frac{\mathbf{z}_p}{\|\mathbf{z}_p\|})}{\sum_{q \in A(i)} \exp(\mathbf{z}_i \cdot \frac{\mathbf{z}_q}{\|\mathbf{z}_q\|})}$$

$$L = L_{CE} + \alpha L_{CL}$$

위와 같은 최종 손실 함수를 통해 모델은 혼합된 멀티모달 특징표현이 원본의 샘플의 고유 정보를 얼마나 포함하는지를 학습한다.

실험 및 결과

- 데이터셋: KEMDy19 Text+Audio 멀티모달 감정인식 데이터셋
중복된 파일, 음성의 발화 세그먼트, 20초 이상의 음성 파일 제거 등의 전처리를 통해 10,214개의 데이터 샘플을 확보
학습 및 평가셋 8:2 무작위 분할
- 방법: 각 모달에 대한 특징표현 기반의 CE, Mixup, CutMix, 제안하는 CutMix-SupCon 방법에 대한 결과 보고

Input	Method	Ace (%)	F1-score (%)
Text	Cross-Entropy	73.42	69.32
Audio		72.59	67.13
Text+Audio	Mixup	73.57	69.08
Text		71.76	67.37
Audio	CutMix	71.46	65.89
Text+Audio		72.39	68.94
Text	CutMix-SupCon (ours)	72.93	69.37
Audio		71.76	65.94
Text+Audio	CutMix-SupCon (ours)	74.74	70.77
Text		72.49	69.72
Audio	CutMix-SupCon (ours)	72.54	68.27
Text+Audio		75.28	71.17

표 1 KEMDy19 데이터셋을 사용한 감정인식 평가 결과

Input	Method	Weighted F1-score(%)
Text+Audio	Cross-Entropy	73.55
	Mixup	72.39
	CutMix	74.77
	CutMix-SupCon (ours)	74.96

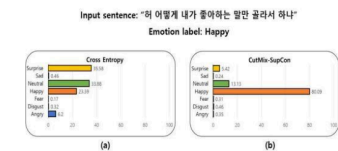


그림 2 KEMDy19 테스트 데이터 샘플에 대한 CE와 CutMix-SupCon의 예측 확률 분포 비교